

2023

Time : 3 hours

Full Marks : 50

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer from **all** the Sections as directed.*

Section – A

(Objective Type Questions)

1. Fill in the blanks : 1×5 = 5
 - (a) For testing $H_0 : \mu = \mu_0$, the suitable test statistics is _____.
 - (b) The degree of freedom of error for R. B. D. is _____.
 - (c) The level of significance is denoted by _____.
 - (d) For $\gamma = 1$, the Student's t-distribution becomes _____.

RA – 4/2

(Turnover)

(e) Three are _____ basic principles of design.

2. Four options are given for each, select the suitable one : $1 \times 5 = 5$

(a) For testing of the independence of two attributes, we use :

- (i) χ^2 (ii) Student's t
(iii) Fisher's t (iv) Snedecor's F

(b) The degree of freedom of error for L.S.D. is :

- (i) $m - 1$ (ii) $m^2 - 1$
(iii) $(m - 1)(m - 2)$ (iv) None of these

(c) The range of χ^2 variate is :

- (i) 0 to ∞ (ii) $-\infty$ to ∞
(iii) $-\infty$ to 0 (iv) None of these

(d) The mean of χ^2 with n degree of freedom is :

- (i) n (ii) $2n$
(iii) $3n$ (iv) None of these

(e) For C.R.D., the experiment units must be :

- (i) Homogeneous (ii) Heterogeneous
(iii) Both (i) and (ii) (iv) None of these

Section – B
(Short-answer Type Questions)

3. Answer any **four** questions of the following :

$$3 \times 4 = 12$$

- (a) Define Latin square design.
- (b) obtain m.g.f. of χ^2 distribution.
- (c) Define χ^2 test and its applications.
- (d) Define Snedecors' F distribution.
- (e) Explain Randomised Block Design (R. B. D.).
- (f) Discuss Unbiased estimator with example.

Section – C
(Long-answer Type Questions)

4. Answer any **four** questions of the following :

$$7 \times 4 = 28$$

- (a) What do you understand about Simple hypothesis, Null hypothesis and Alternative hypothesis.
- (b) Define maximum likelihood estimator. Obtain m.l.e. of the parameter of Poisson distribution.

- (c) Define Student's t statistic and obtain its sampling distribution.
- (d) What is Completely Randomised Design (C.R.D.) ? Give its layout.
- (e) What do you mean by analysis of variance. Give complete analysis of two-way classification.
- (f) Discuss the general procedure of testing the validity of a given hypothesis.

